

Sound Based Computer Automation Using Python

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This project is used to control computer such applications as slideshow, music, or video with a finger snap or a clap. It aims at understanding the automation of computer functions through microphone without using microcontroller boards.

The project uses Python and its package 'sounddevice' to access sound intensity on a microphone inbuilt in a laptop or connected externally. Python package 'pyautogui' is used to automate keystrokes. You may also need to install the PortAudio package by running 'conda install portaudio' on Command Prompt with administrator privileges or by opening Environments through Anaconda Navigator.

A test script is included to test the sensitivity of different microphones and sound ambience. This script plots the intensity of sounds and noises encountered to determine the threshold level of a person's snap/clap. When a finger snap or a clap is detected, a single keystroke is automated.

Operating systems for laptops invariably include/support a voice assistant such as Cortana (MS Windows), Siri (Mac OS), and Mycroft (Linux). Voice assistants automate tasks but are slow and not customisable. This project overcomes all these limitations to automate any task imaginable by a developer.

How it works?

The computer application running in the background can be controlled by automating keystrokes. Fig. 1 gives a brief overview of the procedure. Microphone records sound using Python script. The sound intensity is analysed and keystrokes are mimicked to control applications such as presentation (slideshow), music, or

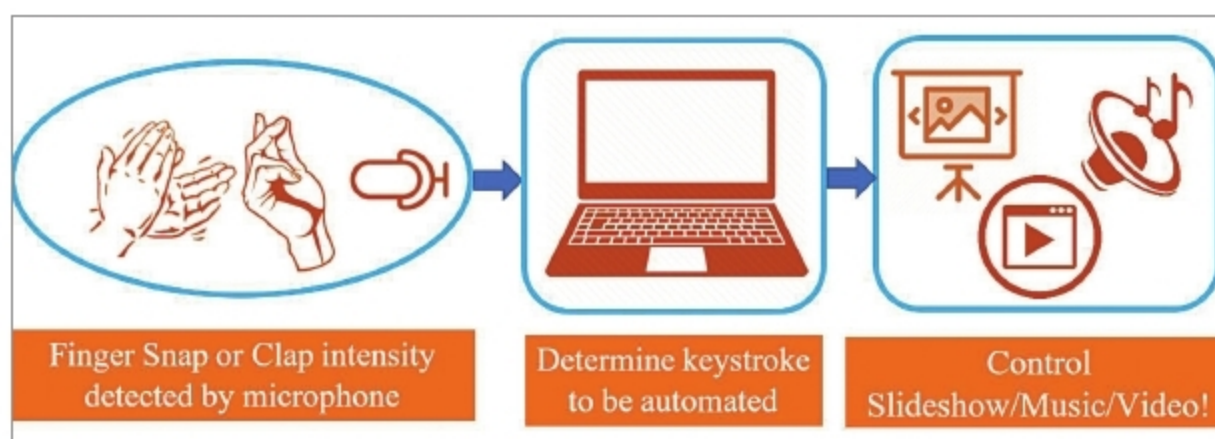


Fig. 1: Block diagram of the project

OS (C:) > AutomateBySound	→ Main folder
AutomateBySound_Main.ipynb	→ Main script for automation
AutomateBySound_Test.ipynb	→ Testing script to determine threshold
PyAutoGUI-0.9.48.tar.gz	→ Source file for installing pyautogui
sounddevice-0.3.14.tar.gz	→ Source file for installing sound device
zReadme_AutomateBySound.txt	→ Information about the project

Fig. 2: Working environment

```

C:\>cd\

C:\>cd AutomateBySound

C:\AutomateBySound>pip install PyAutoGUI-0.9.48.tar.gz

C:\AutomateBySound>pip install sounddevice-0.3.14.tar.gz
  
```

Fig. 3: Installing packages using pip command

video. For example, music is muted or unmuted by mimicking multimedia 'mute' key on the keyboard. The keystrokes can be changed as per requirements.

Creating a work environment

Use Anaconda Distribution on Windows Operating System. Anaconda is preinstalled with Jupyter Notebook, Python 3.xx, and many other Python packages. The AutomateBySound zipped folder consists of five files, as shown in Fig. 2. For simplicity, keep the unzipped AutomateBySound

folder in C drive (location C:\AutomateBySound >).

Software installation and running the project

Step 1. First, install Python 3.7 Distribution from www.anaconda.com. With this, Anaconda PowerShell Prompt, Jupyter Notebook, and other Python packages will also get installed. Then, open Command Prompt. (For running in Windows 7, you would need to run PowerShell from Start Menu on the desktop to open Command Prompt.) Install